

# Thesis presentation: VR Avatars for Older Adults

**By Ivan Pavlov**

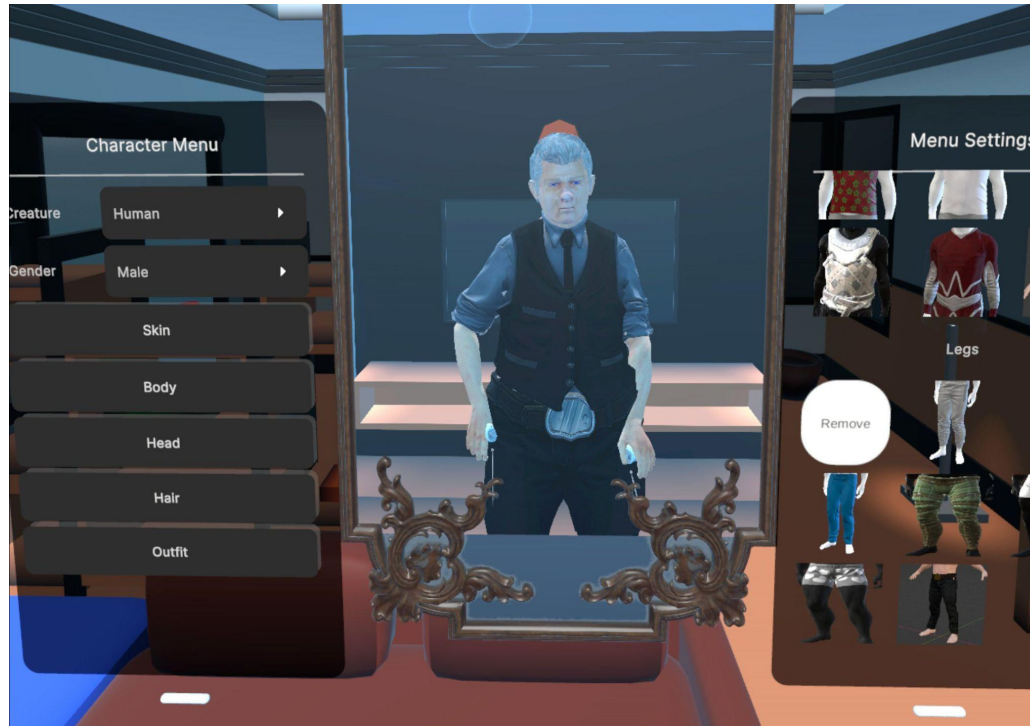
Supervisor: Marvin Wolf

# VR Avatars for Older Adults



A picture from the Internet [1].

# VR Avatars for Older Adults



A screenshot of the resulted VR Avatar Creator prototype.

# Agenda

1. Introduction
2. Research questions
3. Methods
4. Results
5. Limitations
6. Conclusion and Outlook

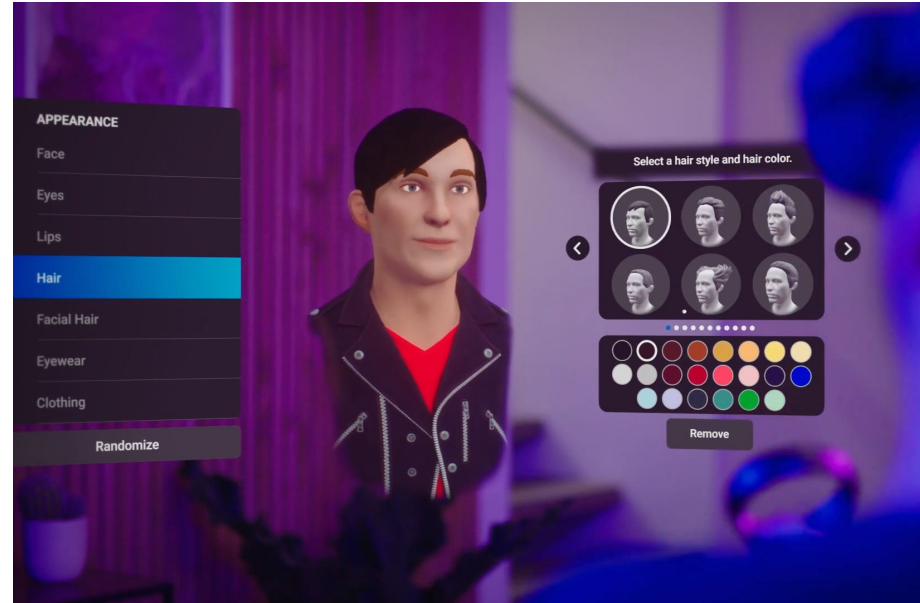
# Introduction

- Virtual reality (VR) provides benefits for older adults [2]
- Avatars can be customized to represent a desired body
- Modern avatar creators focus on the preferences of younger users [3]

# Introduction



A screenshot of *Ready Player Me* [4] avatar creator.



A demonstration of an avatar creation in the Oculus Quest system [5].

# Introduction



A screenshot of an avatar creator *MetaHuman* [6].

# Research Questions

**RQ1** What kind of thoughts and considerations regarding customization options of VR avatars can be found in existing literature?

**RQ2** What would an interactive VR avatar creation tool tailored for older adults look like?

# Methods(1/2) - Related Studies

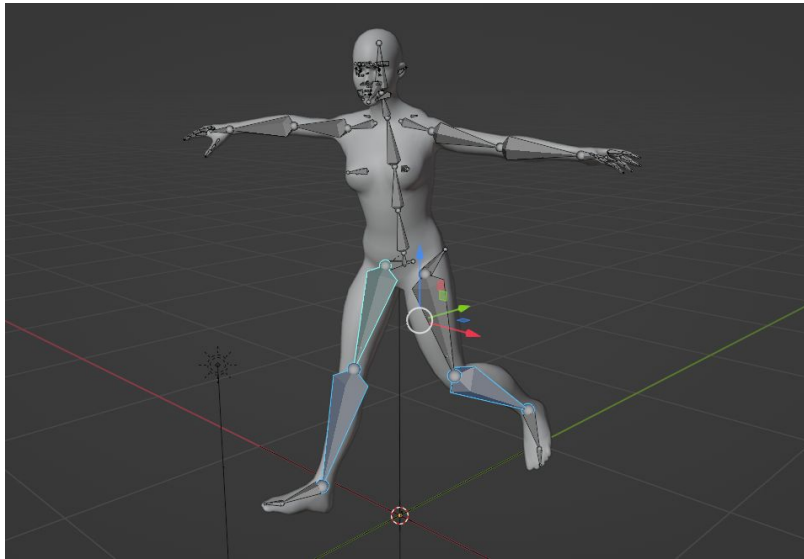
| Authors               | Title                                                                                                                             | Year |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------|------|
| Wei et al. [7]        | Bridging the Generational Gap: Exploring How Virtual Reality Supports Remote Communication Between Grandparents and Grandchildren | 2023 |
| Rice et al. [8]       | Comparing avatar game representation preferences across three age groups                                                          | 2013 |
| Mikhailova et al. [9] | Age and Realism of Avatars in Simulated Augmented Reality: Experimental Evaluation of Anticipated User Experience.                | 2024 |
| Baker et al. [10]     | Exploring the Design of Social VR Experiences with Older Adults.                                                                  | 2019 |
| Carrasco [11]         | Older Adults Designing Avatars for Socializing.                                                                                   | 2018 |
| Puri et al. [12]      | To be (me) or not to be? photorealistic avatars and older adults.                                                                 | 2017 |
| Carrasco et al. [3]   | Negotiating stereotypes of older adults through avatars.                                                                          | 2017 |
| Siriaraya, Ang [13]   | The Social Interaction Experiences of Older People in a 3D Virtual Environment.                                                   | 2019 |
| Cheong et al. [14]    | Avatar: a virtual face for the elderly.                                                                                           | 2011 |

# Methods(1/2) - Accumulating Preferences

1. Collecting statements, feedback and research notes from each study
2. Grouping items by related subjects
3. Identifying and extracting preferences

# Methods(2/2) - Developing VR Prototype

- Unity 2022.3.41f1 [15]
- Unity Multipurpose Avatar ( UMA ) v2.11 [16]



Rigged humanoid model, a screenshot from Blender [17].



Humanoids created and customized with the use of UMA

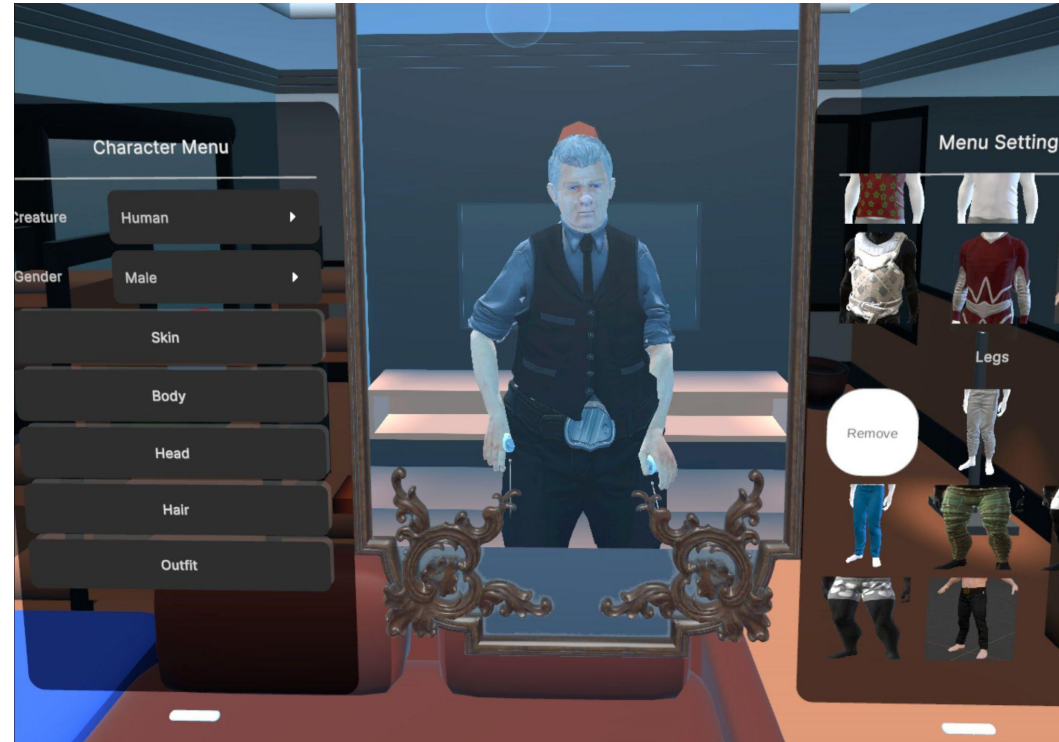
# Results - Accumulated Preferences

| Preference type                             | Subject                 | Example                                                                                                                                                                                                                                                                  |
|---------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Specific choice for a characteristic        | Age                     | Forjador created a 30 years old avatar [9]<br>Nik created a 78-year old avatar [9]                                                                                                                                                                                       |
|                                             | Race                    | Mariam created an avatar with asian ethnicity [9]<br>Klaus created an avatar with average race (30% Asian, African and Caucasian) [9]                                                                                                                                    |
|                                             | Gender                  | Amy created an "androgynous" avatar [9]                                                                                                                                                                                                                                  |
|                                             | Hight                   | Nik created a tall female character( 238 cm) [9]                                                                                                                                                                                                                         |
|                                             | Eyes                    | Gordon created avatar with blue green eye color [9]<br>Red Dog created an avatar looking her grandson that has eye color he could "trust" [9]                                                                                                                            |
|                                             | Hair                    | Participants suggested adding reduced hair of head [30][9][10]                                                                                                                                                                                                           |
|                                             | Beard                   | Participants mentioned that they would like to add a beard [9]                                                                                                                                                                                                           |
|                                             | Torso                   | Keki modified the depth and position of the avatar torso and hip depicting some similarity with his body shape [9]                                                                                                                                                       |
|                                             | Hands                   | Forjador expressed a preference for long fingers [9]                                                                                                                                                                                                                     |
|                                             | Skin                    | Gordon applied 'old skin' texture for his avatarGordon applied 'old skin' texture for his avatar [9]                                                                                                                                                                     |
|                                             | Muscles and Weight      | Sally wanted an avatar with "good physique" [9]<br>Red Dog created an avatar looking his grandson that was "well built" [9]<br>Nik created an avatar with high muscle mass [9]<br>Tony added a low level of 'pregnancy stomach' trying to simulate their body figure [9] |
|                                             | Posture                 | Keki modified the depth and position of the avatar torso and hip depicting some similarity with his body shape [9]                                                                                                                                                       |
|                                             | Outfit                  | "I chose to wear military uniform [in VR] because I always wore it when I was young.." [36]                                                                                                                                                                              |
| General customizability of a characteristic | Face                    | "The majority of the participants emphasized the facial features being the most important part of the avatar in assessing the avatars." [30]                                                                                                                             |
|                                             | Emotions                | Participants expressed need to convey emotions, specifically positive [9][30]                                                                                                                                                                                            |
|                                             | Non-human               | Participants allowed possibly of having a non-human representation for specific context [4][9][32]                                                                                                                                                                       |
|                                             | Outfit                  | Nik was concerned she could not find suitable clothing for her avatar [9]                                                                                                                                                                                                |
| General statements                          | Options for women       | Female participants wanted to have more options for clothes or hairstyles [9][32]                                                                                                                                                                                        |
|                                             | Faithful representation | Some participants criticized that avatars looked younger than the age they had chosen [9]                                                                                                                                                                                |
| Avatar creator specifics                    | Mirror                  | Participant expressed a wish to see their real appearance in order to reconstruct it properly as avatar [9]                                                                                                                                                              |
|                                             | User experience         | Some participants wanted the controls to be simpler and less time-consuming [31]                                                                                                                                                                                         |

**Table 2: The compilation of preferences of older adults accumulated from multiple studies**

# Results - Avatar Creator Prototype

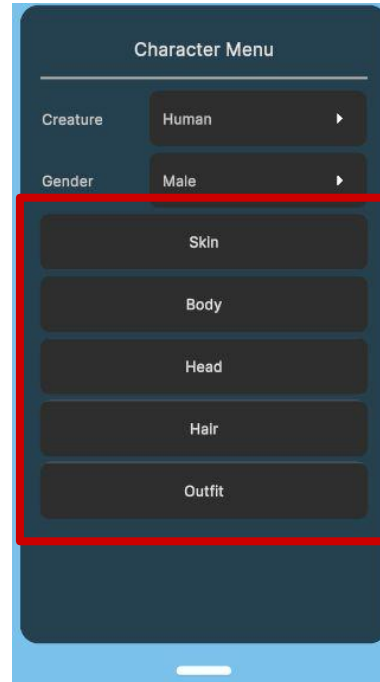
|                    |  |
|--------------------|--|
| Skin               |  |
| Hight              |  |
| Muscles and Weight |  |
| Posture            |  |
| Hands              |  |
| Face               |  |
| Eyes               |  |
| Beard              |  |
| Outfit             |  |
| Age                |  |
| Race               |  |
| Gender             |  |
| Non-human          |  |
| Emotions           |  |



A screenshot of the resulted VR Avatar Creator prototype.

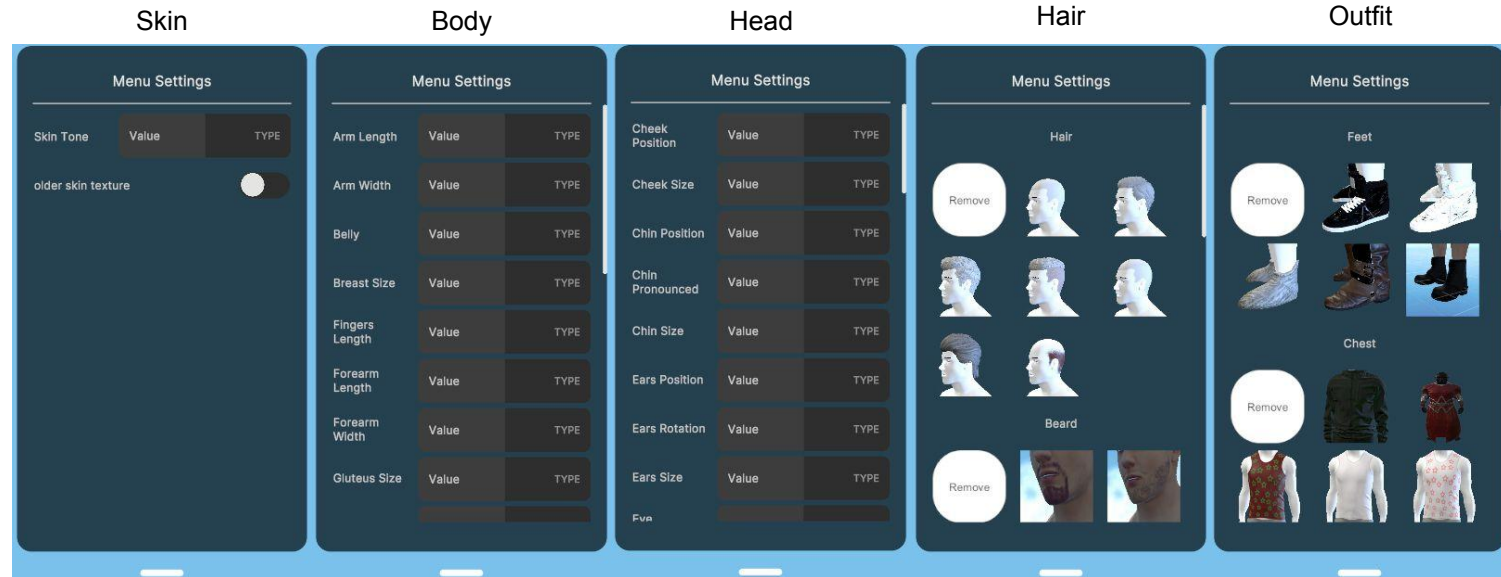
# Results - Avatar Creator Prototype

|                    |  |
|--------------------|--|
| Skin               |  |
| Hight              |  |
| Muscles and Weight |  |
| Posture            |  |
| Hands              |  |
| Face               |  |
| Eyes               |  |
| Beard              |  |
| Outfit             |  |
| Age                |  |
| Race               |  |
| Gender             |  |
| Non-human          |  |
| Emotions           |  |



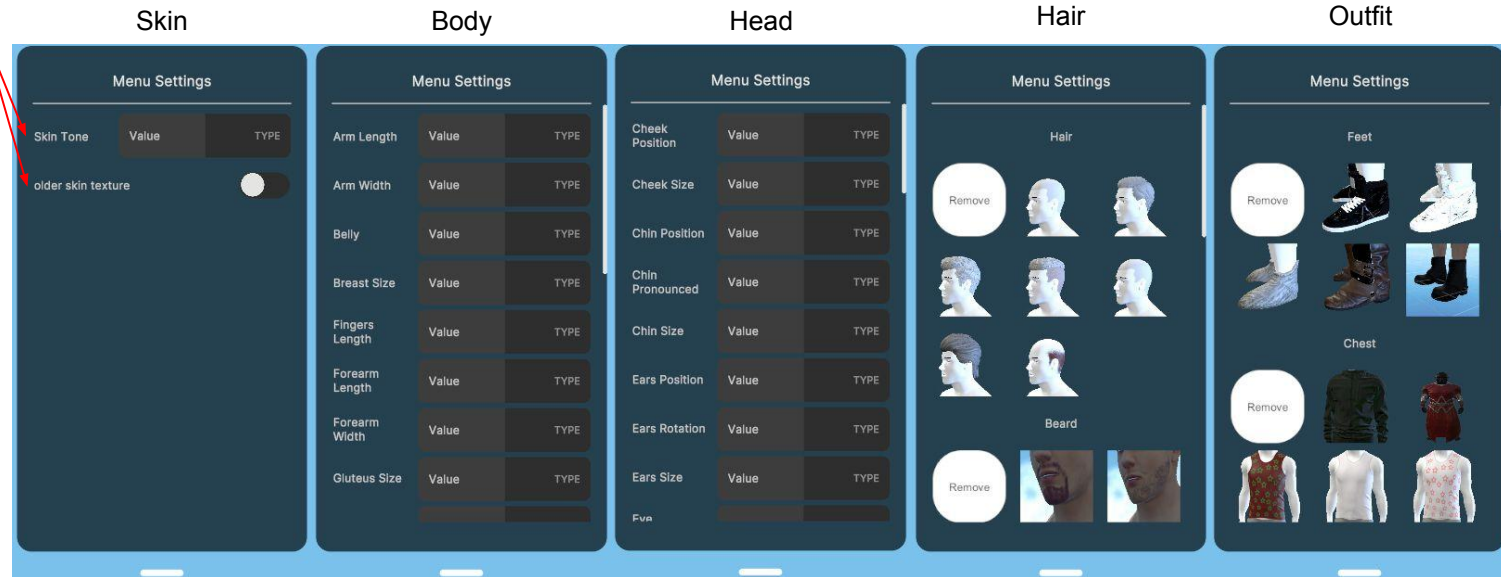
# Results - Avatar Creator Prototype

|                    |  |
|--------------------|--|
| Skin               |  |
| Hight              |  |
| Muscles and Weight |  |
| Posture            |  |
| Hands              |  |
| Face               |  |
| Eyes               |  |
| Beard              |  |
| Outfit             |  |
| Age                |  |
| Race               |  |
| Gender             |  |
| Non-human          |  |
| Emotions           |  |



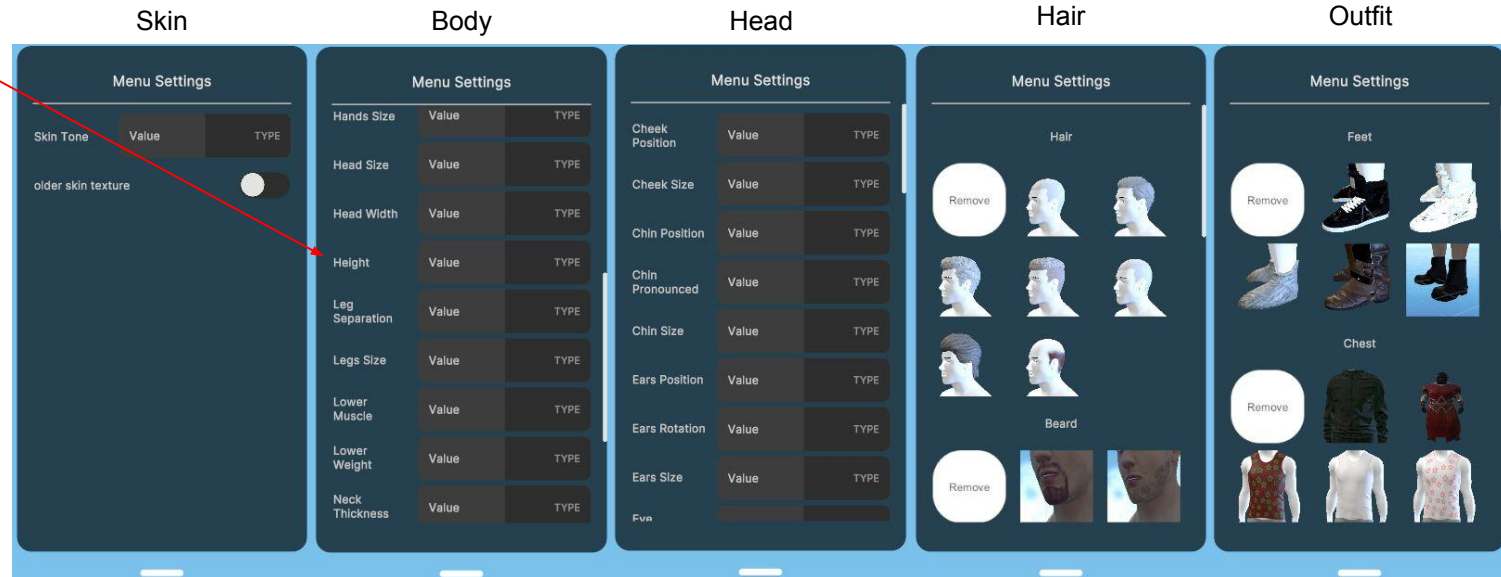
# Results - Avatar Creator Prototype

|                    |                                     |
|--------------------|-------------------------------------|
| Skin               | <input checked="" type="checkbox"/> |
| Height             | <input type="checkbox"/>            |
| Muscles and Weight | <input type="checkbox"/>            |
| Posture            | <input type="checkbox"/>            |
| Hands              | <input type="checkbox"/>            |
| Face               | <input type="checkbox"/>            |
| Eyes               | <input type="checkbox"/>            |
| Beard              | <input type="checkbox"/>            |
| Outfit             | <input type="checkbox"/>            |
| Age                | <input type="checkbox"/>            |
| Race               | <input type="checkbox"/>            |
| Gender             | <input type="checkbox"/>            |
| Non-human          | <input type="checkbox"/>            |
| Emotions           | <input type="checkbox"/>            |



# Results - Avatar Creator Prototype

|                    |   |
|--------------------|---|
| Skin               | ✓ |
| Hight              | ✓ |
| Muscles and Weight |   |
| Posture            |   |
| Hands              |   |
| Face               |   |
| Eyes               |   |
| Beard              |   |
| Outfit             |   |
| Age                |   |
| Race               |   |
| Gender             |   |
| Non-human          |   |
| Emotions           |   |



# Results - Avatar Creator Prototype

|                    |   |
|--------------------|---|
| Skin               | ✓ |
| Height             | ✓ |
| Muscles and Weight | ✓ |
| Posture            |   |
| Hands              |   |
| Face               |   |
| Eyes               |   |
| Beard              |   |
| Outfit             |   |
| Age                |   |
| Race               |   |
| Gender             |   |
| Non-human          |   |
| Emotions           |   |



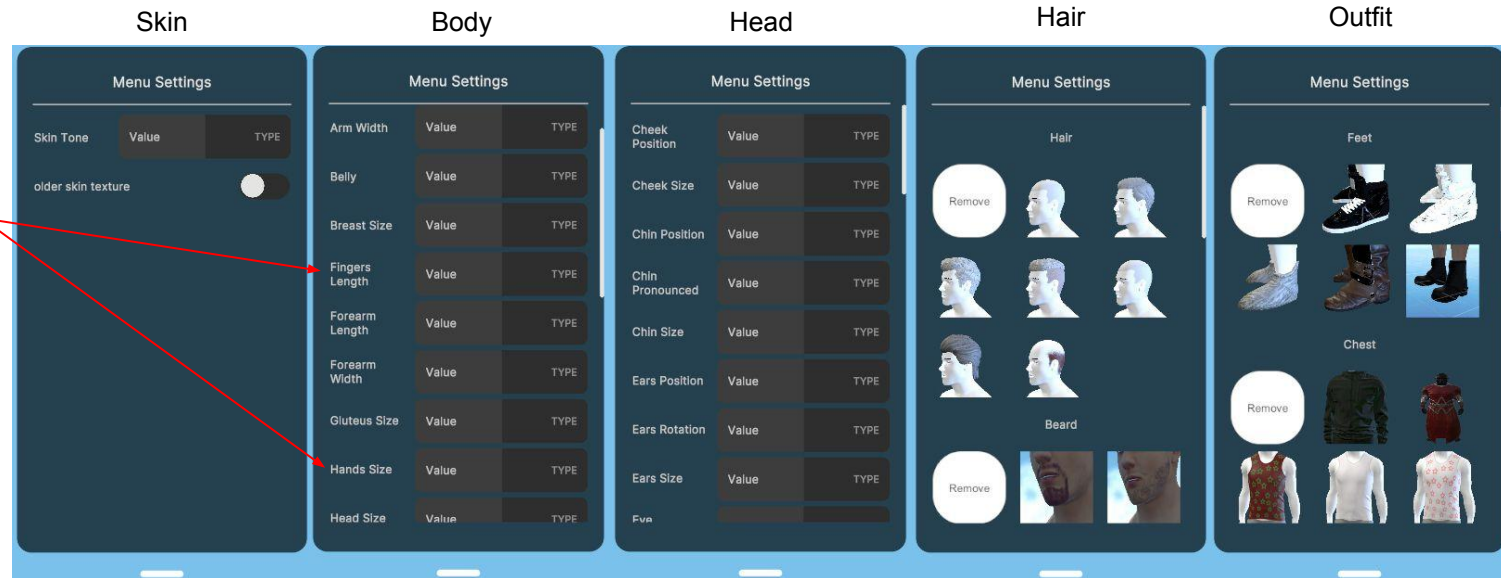
# Results - Avatar Creator Prototype

|                    |   |
|--------------------|---|
| Skin               | ✓ |
| Hight              | ✓ |
| Muscles and Weight | ✓ |
| Posture            | ✓ |
| Hands              |   |
| Face               |   |
| Eyes               |   |
| Beard              |   |
| Outfit             |   |
| Age                |   |
| Race               |   |
| Gender             |   |
| Non-human          |   |
| Emotions           |   |



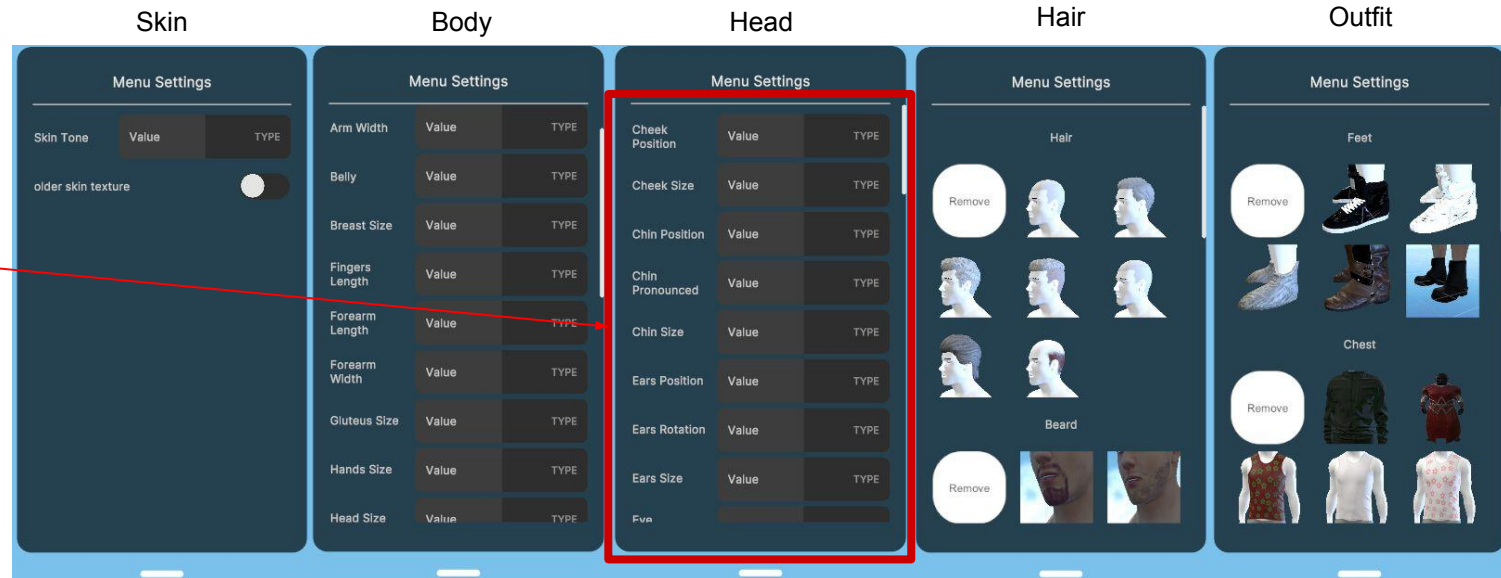
# Results - Avatar Creator Prototype

|                    |   |
|--------------------|---|
| Skin               | ✓ |
| Height             | ✓ |
| Muscles and Weight | ✓ |
| Posture            | ✓ |
| Hands              | ✓ |
| Face               |   |
| Eyes               |   |
| Beard              |   |
| Outfit             |   |
| Age                |   |
| Race               |   |
| Gender             |   |
| Non-human          |   |
| Emotions           |   |



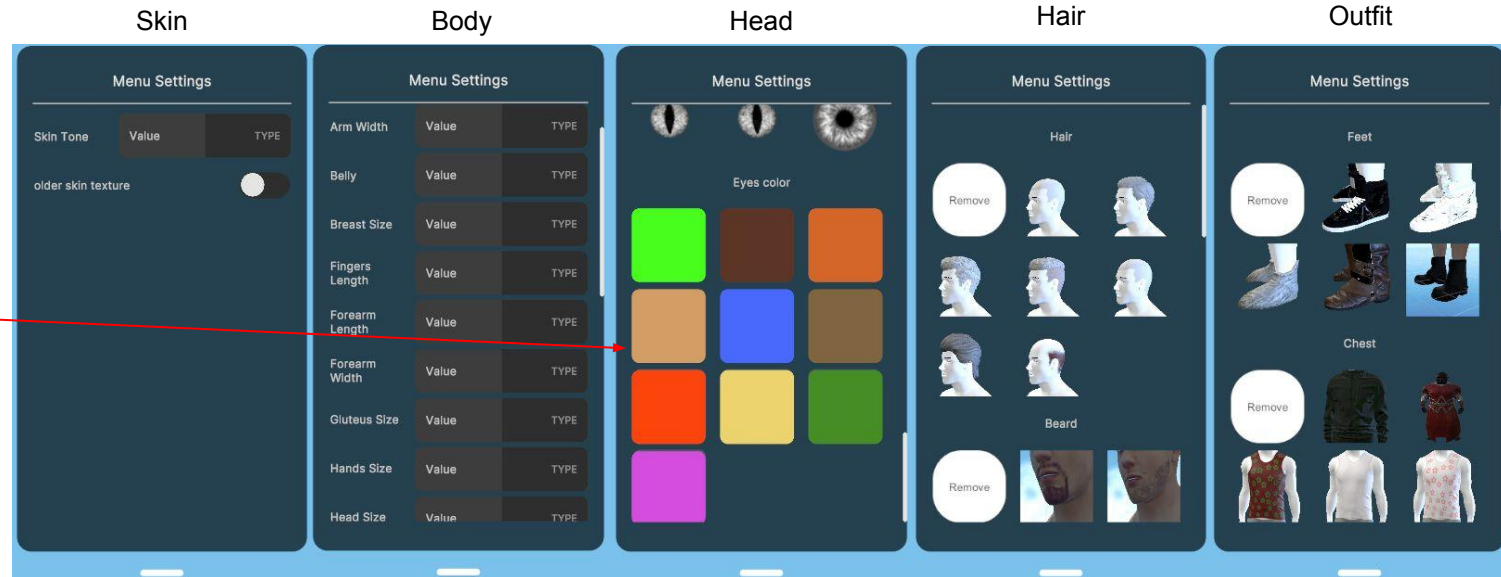
# Results - Avatar Creator Prototype

|                    |   |
|--------------------|---|
| Skin               | ✓ |
| Height             | ✓ |
| Muscles and Weight | ✓ |
| Posture            | ✓ |
| Hands              | ✓ |
| Face               | ✓ |
| Eyes               |   |
| Beard              |   |
| Outfit             |   |
| Age                |   |
| Race               |   |
| Gender             |   |
| Non-human          |   |
| Emotions           |   |



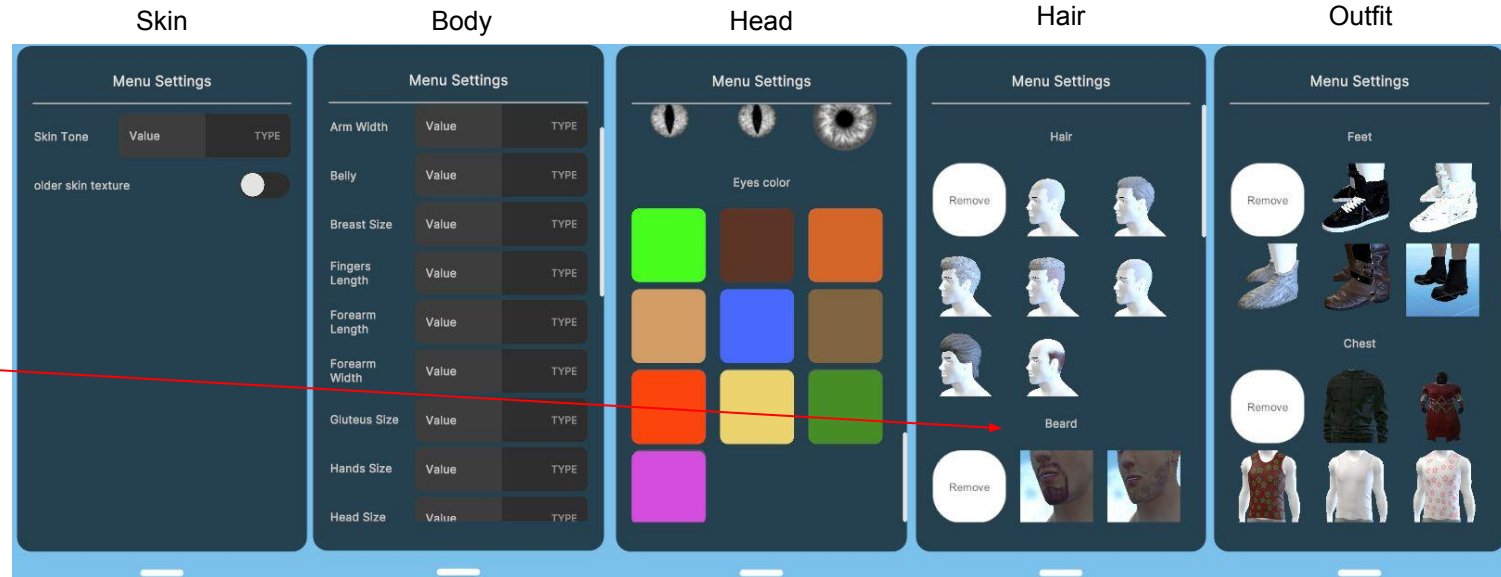
# Results - Avatar Creator Prototype

|                    |   |
|--------------------|---|
| Skin               | ✓ |
| Height             | ✓ |
| Muscles and Weight | ✓ |
| Posture            | ✓ |
| Hands              | ✓ |
| Face               | ✓ |
| Eyes               | ✓ |
| Beard              |   |
| Outfit             |   |
| Age                |   |
| Race               |   |
| Gender             |   |
| Non-human          |   |
| Emotions           |   |



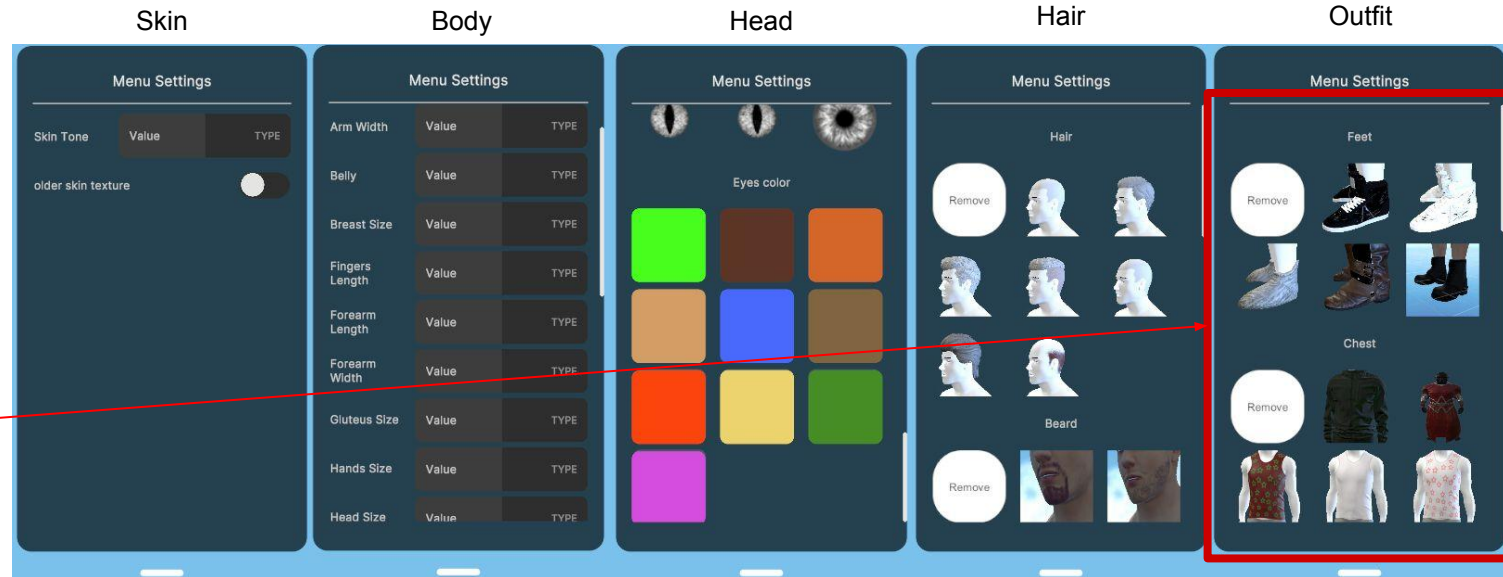
# Results - Avatar Creator Prototype

|                    |   |
|--------------------|---|
| Skin               | ✓ |
| Height             | ✓ |
| Muscles and Weight | ✓ |
| Posture            | ✓ |
| Hands              | ✓ |
| Face               | ✓ |
| Eyes               | ✓ |
| Beard              | ✓ |
| Outfit             |   |
| Age                |   |
| Race               |   |
| Gender             |   |
| Non-human          |   |
| Emotions           |   |



# Results - Avatar Creator Prototype

|                    |   |
|--------------------|---|
| Skin               | ✓ |
| Height             | ✓ |
| Muscles and Weight | ✓ |
| Posture            | ✓ |
| Hands              | ✓ |
| Face               | ✓ |
| Eyes               | ✓ |
| Beard              | ✓ |
| Outfit             | ✓ |
| Age                |   |
| Race               |   |
| Gender             |   |
| Non-human          |   |
| Emotions           |   |



# Results - Avatar Creator Prototype

|                    |   |
|--------------------|---|
| Skin               | ✓ |
| Height             | ✓ |
| Muscles and Weight | ✓ |
| Posture            | ✓ |
| Hands              | ✓ |
| Face               | ✓ |
| Eyes               | ✓ |
| Beard              | ✓ |
| Outfit             | ✓ |
| Age                | ✓ |
| Race               | ✓ |
| Gender             |   |
| Non-human          |   |
| Emotions           |   |



# Results - Avatar Creator Prototype

|                    |   |
|--------------------|---|
| Skin               | ✓ |
| Hight              | ✓ |
| Muscles and Weight | ✓ |
| Posture            | ✓ |
| Hands              | ✓ |
| Face               | ✓ |
| Eyes               | ✓ |
| Beard              | ✓ |
| Outfit             | ✓ |
| Age                | ✓ |
| Race               | ✓ |
| Gender             | ✓ |
| Non-human          | ✓ |
| Emotions           |   |



# Results - Avatar Creator Prototype



Video demonstration of the VR Avatar Creator Prototype.

# Limitations

- Preferences from the literature
  - studies' count
  - participants' count
- Testing and Evaluation by older adults
- Accessibility
  - Sliders, UI
- Not complete features
  - skin, genders, outfits
  - emotions
  - animations

# Conclusion & Outlook

- Investigated preferences of older adults
- Developed a prototype of a VR avatar creator
- In the future: evaluation with target demographic

# References(1/2)

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# Backup Slides



A Meme from the Internet [18].

# Skin texture

- Constructed from a real head scan of an older person [19] [20]
- Image-edited to match original UV of the UMA texture
- Used as an overlay on top of the original texture on the model

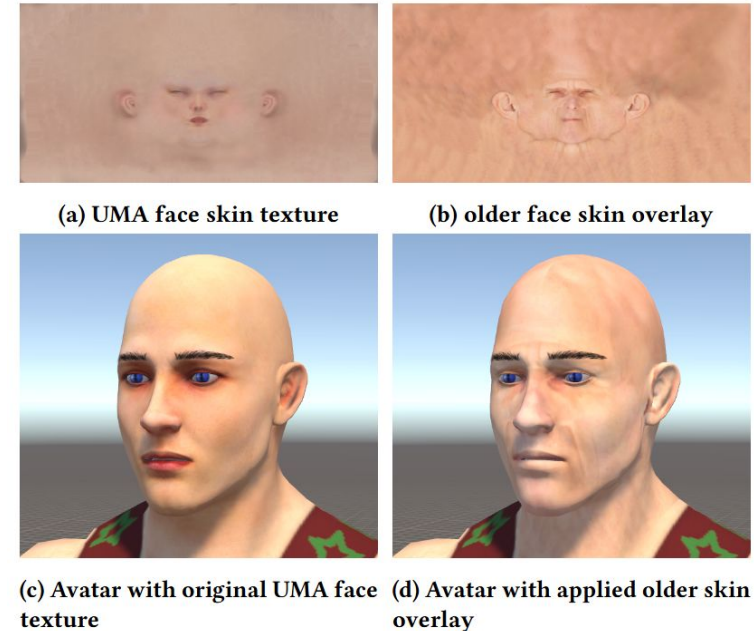


Figure 4: Applying skin texture customisation with UMA

# VR Embodiment

- Translates the user's controller inputs into the virtual world depending on the virtual body characteristics
- longer virtual arms means wider reach
- Requires calibration for personalised results ( T-Pose)

